

Deep Learning-Based Cell Segmentation Pipeline for Analyzing Cancer Cell Migration Patterns

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Background

Cell migration plays a crucial role in cancer metastasis, which remains the leading cause of mortality among cancer patients. Current research, as demonstrated in the paper by Tagay et al. [1], shows that cancer cells exhibit complex migration patterns guided by both actomyosin and dynein-powered locomotion mechanisms. To better understand these migration patterns, accurate cell detection and tracking are essential. However, the first fundamental step is developing a robust cell segmentation method that can precisely identify cell boundaries in microscopy images.

Importance

Accurate cell segmentation is crucial for analyzing cell morphology changes during migration. Automated segmentation reduces manual analysis time and eliminates human bias. Better segmentation leads to more reliable cell tracking and motion analysis. Understanding cell migration patterns is key to developing anti-metastatic therapies. The pipeline will serve as a foundation for future work on cell tracking algorithms. Results could provide quantitative insights into cancer cell behavior under different conditions.

Approach

The project will benefit from pretrained U-Net based deep learning architectures for semantic segmentation of cells in microscopy images. The pipeline will begin with dataset preparation, including preprocessing phase contrast microscopy images and creating automatic annotation pipeline for training. The U-Net architecture will be modified to better handle phase contrast images, potentially incorporating attention mechanisms to improve feature extraction. Training may involve optimization of hyperparameters, implementation of appropriate loss functions, and use of cross-

validation to ensure model robustness. Model performance will be evaluated using standard metrics such as IoU and Dice coefficient, with comparisons to other segmentation methods.

Future work will focus on integrating the segmentation model with tracking algorithms to analyze cell motion patterns. This will include developing visualization tools and potentially extending the pipeline to handle 3D cell migration data.

References:

[1] Tagay Y, Kheirabadi S, Ataie Z, Singh RK, Prince O, Nguyen A, Zhovmer AS, Ma X, Sheikhi A, Tsygankov D, Tabdanov ED. Dynein-Powered Cell Locomotion Guides Metastasis of Breast Cancer. Advanced Science. 2023 Nov;10(31):2302229.